Extra Bite #4: MMICs and RF Component Packaging

Why GaAs and GaN beat silicon at microwaves, and why through-hole parts go sideways above 300 MHz.

Group E6E · 12 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Why You Care

At HF, the parts you solder onto a board are still acting like the symbol on the schematic. As frequency climbs into UHF and microwave, the package itself starts dominating behavior – lead lengths become antennas, parasitic capacitance shunts signal to ground, and the chemistry of the semiconductor matters more than the schematic does. E6E tests recognition of **MMICs** (Monolithic Microwave Integrated Circuits), the **materials** that make them work at high frequency, and the **packages** that get used at RF vs the ones that don't.

Three Materials, Three Speed Tiers

Material	Best for	Why
Silicon (Si)	HF, VHF, low UHF	Workhorse. Cheap, easy to fab, plenty of speed up to about 1 GHz.
Gallium arsenide (GaAs)	UHF and higher	Higher **electron mobility** than silicon – electrons move faster through the crystal, so the device switches faster. (E6E01)
Gallium nitride (GaN)	Highest-frequency MMICs (mm-wave)	Supports the highest frequencies in MMICs. Wide bandgap and high breakdown voltage let it sustain high power at millimeter-wave frequencies where GaAs would fail. (E6E03)

MMIC Basics

An MMIC is a complete RF amplifier (or mixer, etc.) built on a single semiconductor die, designed to drop into a 50-ohm RF system with minimal external matching.

Question	Answer	Q
Most common input AND output impedance	**50 ohms** (matches a normal RF system)	E6E04
Typical noise figure of a low-noise UHF preamp	About **0.5 dB**	E6E05
What makes MMICs popular for VHF/ microwave?	Controlled gain, low noise figure, constant input/output impedance over the rated frequency range	E6E06
Transmission line used to connect to MMICs on a PCB	**Microstrip** (a strip of copper over a ground plane – behaves like a controlled 50-ohm line)	E6E07
How is DC power supplied to a typical MMIC?	Through a **resistor and/or RF choke** connected to the amplifier's **output** lead. The choke blocks RF from leaking into the supply; the resistor sets the bias current.	E6E08

Packaging: Through-Hole vs Surface-Mount

The exam wants you to know the difference and why it matters at RF.

Package	Style	RF behavior
DIP (Dual In-line Package)	Through-hole: **two rows of pins on opposite sides**, plug into holes in the board. (E6E11)	Long pin lengths add inductance. **Excessive lead length** is why DIPs are not used at UHF and higher. (E6E12) DIP is itself a through-hole type. (E6E02)
Surface-mount (SMD/SMT)	Tiny package with leads or pads that solder directly to the PCB surface – no holes.	Lowest parasitic effects above HF. (E6E09) Smaller area, shorter traces, less parasitic L and C – **all of the above**. (E6E10)

Mental Cheat Sheet

- GaAs → useful at UHF+ because of higher electron mobility.
- GaN → highest-frequency MMIC material.
- MMIC standard input/output Z = **50 ohms**.
- Low-noise UHF preamp noise figure: about **0.5 dB**.
- MMICs popular: controlled gain, low NF, constant Z over the band.
- Microstrip is the PCB transmission line to MMICs.
- MMIC DC power: **resistor and/or RF choke on the output lead**.
- DIP = Dual In-line Package: **two rows of pins on opposite sides**, through-hole.
- DIP fails at UHF because of **excessive lead length**.
- Surface-mount has the least parasitic effect above HF.

Common Traps

- GaAs is faster than silicon specifically because **electron mobility** is higher – not any "lower junction voltage" or "lower noise figure" trick.
- MMIC bias goes in on the **output** lead through a resistor + RF choke, not the input lead.
- DIP fails at UHF because of **lead length** (parasitic inductance), not anything about dielectrics or epoxy.
- "All these choices are correct" is the right answer for E6E10 (surface-mount advantages) – do not dismiss it as a cop-out option.
- MMIC input AND output Z are both **50 ohms**. Trap answers: 75, 300, 450.

- Surface-mount vs through-hole = smaller + shorter + less L/C (**all of the above**).

Takeaway: Materials get faster as you go silicon → GaAs → GaN; MMICs are 50-ohm building blocks fed power through a choke on the output and connected by microstrip; surface-mount won the high-frequency packaging fight because through-hole pins look like inductors at UHF.

Bite 4 Question Bank (E6E)

12 questions. Read, decide, check the answer key at the end.

E6E01. Why is gallium arsenide (GaAs) useful for semiconductor devices operating at UHF and higher frequencies?
A. Higher noise figures.
B. Higher electron mobility.
C. Lower junction voltage drop.
D. Lower transconductance.

E6E02. Which of the following device packages is a through-hole type?
A. DIP.
B. PLCC.
C. BGA.
D. SOT.

E6E03. Which of the following materials supports the highest frequency of operation when used in MMICs?
A. Silicon.
B. Silicon nitride.
C. Silicon dioxide.
D. Gallium nitride.

E6E04. Which is the most common input and output impedance of MMICs?
A. 50 ohms.
B. 300 ohms.
C. 450 ohms.
D. 75 ohms.

E6E05. Which of the following noise figure values is typical of a low-noise UHF preamplifier?
A. 0.5 dB.
B. −10 dB.
C. 44 dBm.
D. −20 dBm.

E6E06. What characteristics of MMICs make them a popular choice for VHF through microwave circuits?
A. The ability to retrieve information from a single signal, even in the presence of other strong signals.
B. Extremely high Q factor and high stability over a wide temperature range.
C. Nearly infinite gain, very high input impedance, and very low output impedance.
D. Controlled gain, low noise figure, and constant input and output impedance over the specified frequency range.

E6E07. What type of transmission line is often used for connections to MMICs?
A. Miniature coax.
B. Circular waveguide.
C. Parallel wire.
D. Microstrip.

E6E08. How is power supplied to the most common type of MMIC?
A. Through a capacitor and RF choke connected to the amplifier input lead.
B. MMICs require no operating bias.
C. Through a resistor and/or RF choke connected to the amplifier output lead.
D. Directly to the bias voltage (Vcc) lead.

E6E09. Which of the following component package types have the least parasitic effects at frequencies above the HF range?
A. TO-220.
B. Axial lead.
C. Radial lead.
D. Surface mount.

E6E10. What advantage does surface-mount technology offer at RF compared to using through-hole components?
A. Smaller circuit area.
B. Shorter circuit board traces.
C. Components have less parasitic inductance and capacitance.
D. All these choices are correct.

E6E11. What is a characteristic of DIP packaging used for integrated circuits?
A. Extremely low stray capacitance (dielectrically isolated package).
B. Extremely high resistance between pins (doubly insulated package).
C. Two chips in each package (dual in package).
D. Two rows of connecting pins on opposite sides of package (dual in-line package).

E6E12. Why are DIP through-hole package ICs not typically used at UHF and higher frequencies?
A. Excessive dielectric loss.
B. Epoxy coating is conductive above 300 MHz.
C. Excessive lead length.
D. Unsuitable for combining analog and digital signals.

Answer Key

E6E01: B	E6E02: A	E6E03: D
E6E04: A	E6E05: A	E6E06: D
E6E07: D	E6E08: C	E6E09: D
E6E10: D	E6E11: D	E6E12: C